Adi Karthikeya Sunkara Balasubramanyam

Data Scientist | **AI/ML Engineer** | **Data Analyst** | **Data Engineer**
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Professional Summary

M.S. Data Science graduate building ML models, RAG/agentic AI, and pipelines in Python/SQL; Databricks and Claude certified. Spans classical ML, LLM applications, and BI dashboarding, using Claude Code daily.

Experience

Technical Intern Sep 2026 - Present
Worldlink *Frisco, TX*
- Collaborate with senior data scientists and project managers to design and deliver technical solutions, supporting data preparation, model testing, and performance evaluation across AI/ML projects in Python.
- Build automation scripts and monitoring dashboards; manage code and documentation across team repositories.
- Document findings and present progress in daily stand-ups and weekly leadership reviews, using Jira and Office 365.

Teaching Assistant Jan 2025 - May 2026
University of Houston *Houston, TX*
- Supported 120+ students in Pattern Recognition and Machine Learning and Engineering Data Analytics courses.
- Led debugging sessions on Python, SQL, and scikit-learn workflows: EDA, feature selection, and model evaluation.
- Reworked assignment rubrics for CEA 4341 to speed up grading turnaround across the cohort.

Data Science Intern, NSF Data Science for Energy Transition (Federally Funded) May 2025 - Jun 2025
University of Houston *Houston, TX / Remote*
- Selected for a nationally funded 5-week NSF program in data science for energy and geoscience; profiled, cleaned, and transformed multi-source geophysics and energy datasets alongside researchers and industry professionals.
- Built facies classification models (KNN, Random Forest, Decision Tree) on well-log data, scoring 0.8024 accuracy in a national Kaggle-style competition.
- Applied PCA and K-Means to structure raw Tellus geophysics data into interpretable subsurface terranes, and built Tableau dashboards and QGIS maps converting raw EIA and Census records into decision-ready reporting for stakeholders.

Certifications

Databricks Certified Generative AI Engineer Associate, Oct 2026 | **Claude Certified Architect - Foundations**, Sep 2026

Skills

Data Engineering: SQL, PostgreSQL, ETL/ELT, Data Pipelines, Data Ingestion, Data Transformation, Data Profiling, Data Quality & Validation, Source-to-Target Mapping, Data Modeling, Window Functions, Query Optimization, Neo4j, ChromaDB
Analytics & BI: Tableau, Streamlit, Excel, QGIS, Exploratory Data Analysis, KPI Definition, Hypothesis Testing, Statistical Analysis, Segmentation, Dashboarding, Stakeholder Reporting
Machine Learning: Scikit-learn, XGBoost, LightGBM, TensorFlow, SHAP, Feature Engineering, Ensemble Models, Classification, Clustering, Model Evaluation & Calibration, PCA, K-Means
AI & NLP: Claude, Gemini, Groq, LangGraph, LangChain, Google ADK, Agentic Workflows, RAG, Vector Search, Embeddings, Prompt Engineering, Transformers, BERT, Databricks (Vector Search, Model Serving, Unity Catalog), MLflow, LLM Chains
Languages & Tools: Python, SQL, R, Pandas, NumPy, Git, Docker, FastAPI, Vercel, Render

Projects

RareDx, Agentic Rare Disease Diagnostic Assistant | *LangGraph, Neo4j, ChromaDB, Groq* Mar 2026 - May 2026
- Built an ingestion and enrichment pipeline loading full Orphanet disease profiles into a ChromaDB vector store with BioLORD embeddings, lifting retrieval Top-1 accuracy from 10% to 92.5% and Recall@5 to 100% across 40 adversarial cases.
- Designed a validation and calibration layer over 7 retrieval strategies (280 runs, Brier 0.068, ECE 0.051), zero hallucination.

Predictive Match ML Dashboard | *XGBoost, LightGBM, SHAP, Streamlit, SQL/Pandas* Jun 2026
- Engineered a 52-feature transformation pipeline over 60,000 user profiles, trained stacked ensemble models, and shipped a Streamlit analytics dashboard using SHAP explainability to rank personalized, data-backed improvements.

ShopperAI, Agentic AI Shopping Assistant | *Python, Google ADK, Gemini, LangChain* Dec 2025
- Built a deterministic, stage-based agentic shopping assistant guiding users through product discovery and virtual try-on, ranking 4th globally out of 200 teams at the ODSC AI & Google Cloud Agentic AI Hackathon.

Question Answering System (NLP) | *BERT, RoBERTa, DeBERTa, Flan-T5* Sep 2025 - Dec 2025
- Fine-tuned four transformer architectures on SQuAD v2, improving Exact Match from 34.9% to 75.3% and F1 up to 78.4%.

Olist E-Commerce Analytics | *PostgreSQL, SQL, Tableau* May 2026
- Modeled 100K orders in PostgreSQL with 9 SQL queries (CTEs, window functions, cohorts) and a Tableau dashboard; found late delivery (8+ days) drops reviews from 4.3 to 1.7 stars.

Education

University of Houston Houston, TX
M.S. Data Science, GPA 3.9/4.0 *May 2026*

Amrita Vishwa Vidyapeetham Bengaluru, India
B.Tech, Electronics and Communication Engineering, GPA 3.50/4.00 *May 2024*